

Task 1: Normal Equation in Linear Regression

1. Definition of the Normal Equation

The Normal Equation is an analytical method used to find the optimal parameters (weights) of a Linear Regression model in a single step. Unlike iterative optimization techniques such as Gradient Descent, the Normal Equation solves for the global minimum of the Ordinary Least Squares (OLS) cost function algebraically without requiring multiple epochs, iterations, or a learning rate.

2. Mathematical Formulation and Derivation

Consider a linear regression problem with m training samples and n features. Let X be the feature matrix of size $m \times (n + 1)$, where the first column is populated with 1s to account for the intercept term, and let y be the target vector of size $m \times 1$. The hypothesis function is expressed in matrix form as:

$$\hat{y} = X\theta$$

where θ is the parameter vector of size $(n + 1) \times 1$.

The Sum of Squared Errors cost function $J(\theta)$ is defined as:

$$J(\theta) = (1 / 2) \|X\theta - y\|^2 = (1 / 2) (X\theta - y)^T (X\theta - y)$$

Expanding the matrix product:

$$J(\theta) = (1 / 2) (\theta^T X^T X \theta - 2 (X^T y)^T \theta + y^T y)$$

To find the value of θ that minimizes $J(\theta)$, we take the partial derivative with respect to θ and set the resulting gradient to zero:

$$\nabla_{\theta} J(\theta) = X^T X \theta - X^T y = 0$$

$$X^T X \theta = X^T y$$

Assuming that the matrix $(X^T X)$ is invertible (non-singular), we multiply both sides by $(X^T X)^{-1}$ to obtain the Normal Equation:

$$\theta = (X^T X)^{-1} X^T y$$

3. Optimal Parameter Estimation

The resulting vector θ gives the exact weights that minimize the residual sum of squares between the predictions and the actual target values. Because the quadratic cost function $J(\theta)$ is strictly convex, setting the gradient to zero is mathematically guaranteed to produce the global minimum. If $(X^T X)$ is not invertible (due to collinear features or when $m < n$), the Moore-Penrose pseudo-inverse can be used instead: $\theta = (X^T X)^+ X^T y$.

4. Advantages and Disadvantages vs. Gradient Descent

Advantages of the Normal Equation:

- No need to choose or tune a learning rate (α).
- No iterative steps or convergence checks; computes exact analytical solution directly.
- Feature scaling (normalization or standardization) is not required.

Disadvantages of the Normal Equation:

- Computationally expensive: computing the matrix inverse $(X^T X)^{-1}$ requires $O(n^3)$ time complexity, making it slow when the number of features n is very large (e.g., $n > 10,000$).
- Inefficient for extremely large datasets where Gradient Descent (especially Mini-batch or Stochastic GD) scales much better.
- Potential issues with matrix singularity if redundant or linearly dependent features exist.

Dimension	Normal Equation	Gradient Descent
Computational Complexity	$O(n^3)$	$O(k \cdot n \cdot m)$
Hyperparameters	None	Requires tuning learning rate (α) and epochs
Process	Analytical (1 step)	Iterative (many steps)
Feature Scaling	Not required	Required / Highly recommended
Scalability	Best for small/moderate n	Suitable for large-scale datasets